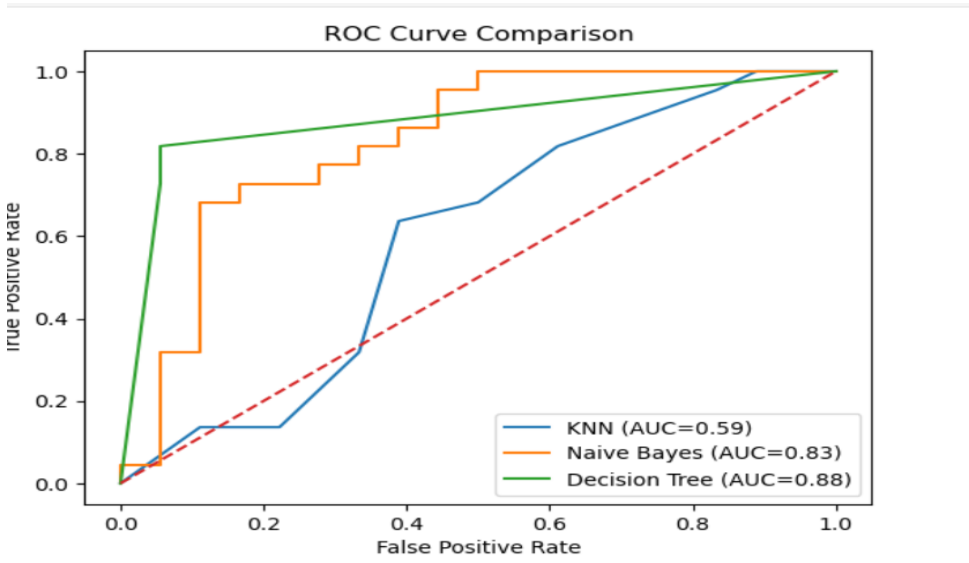


Model Comparison Report: Delivery Status Prediction

This report compares the performance of three machine learning classifiers—KNN, Naive Bayes, and Decision Tree—for predicting delivery status. The models are evaluated using Accuracy, Precision, Recall, and F1-score. ROC curve analysis is also used to assess the classification capability.

Model	Accuracy	Precision	Recall	F1 Score
KNN	0.600	0.625	0.682	0.652
Naive Bayes	0.750	0.750	0.818	0.783
Decision Tree	0.875	0.947	0.818	0.878

Performance Comparison: Decision Tree achieves the highest accuracy (87.5%) and precision (0.947), indicating superior prediction performance. Naive Bayes provides balanced results with moderate accuracy and good recall. KNN performs the weakest with the lowest accuracy and F1-score.



Actionable Insights: KNN: Simple but lower predictive performance. Naive Bayes: Balanced and computationally efficient. Decision Tree: Best overall performance with strong interpretability. Recommendation: The Decision Tree classifier is recommended for predicting delivery status because it achieves the highest accuracy, precision, and F1-score while maintaining interpretability.